



# Latency, Availability & Rate Limits

Week 4 • Day 4 • Technical Architecture & Constraints

TPM Academy



# Day 4 Objectives

- Distinguish **SLI / SLO / SLA / error budget**
- Set the **three SLOs**: latency, availability, throughput / rate limit
- Walk the **latency budget** across your Container diagram — does the math fit?
- Design a **rate-limit policy** with explicit failure-mode behavior
- Name an **error-budget consequence** — the behavior change when we breach



# Today's Shape

| Clock         | Block                       | Outcome                                  |
|---------------|-----------------------------|------------------------------------------|
| 09:00 – 09:15 | Opening                     | Pin Container diagram; recap SLI/SLO/SLA |
| 09:15 – 10:30 | Block 1 + Activity 1        | SLO Triage                               |
| 10:45 – 12:00 | Block 2 + Activity 2        | Set your three SLOs                      |
| 13:00 – 14:15 | Block 3 + Activity 3        | Latency-budget walk                      |
| 14:30 – 16:00 | Block 4 + Activity 4 + Wrap | Cross-review + AI sanity check           |

# Block 1 — The Vocabulary

SLI / SLO / SLA / Error budget



# Get the Words Right

| Term         | Definition                                     | Audience              |
|--------------|------------------------------------------------|-----------------------|
| SLI          | Measurement: latency, availability, error rate | Operators / engineers |
| SLO          | Target for the SLI                             | TPMs, eng leads       |
| SLA          | Contractual commitment with consequences       | Customer-facing       |
| Error budget | $(1 - \text{SLO})$ over the window             | Eng leads, TPMs       |

**Common error:** "We commit to 99.99%" written casually = SLA, not SLO. Use SLO for internal targets; SLA only when contractually intended.

# Three SLOs Are Usually Enough

## Latency

" $p \leq ms$  % of the time over "

Anchor: when does the user give up?

## Availability

"% of requests succeed over "

Anchor: cost of a minute of downtime?

## Throughput / rate limit

"system handles rps per "

Anchor: what abuse / accident are we protecting against?

**Each must have:** a percentile, a window, and a defense.



# SLO Failure Modes

| Failure                       | Example                                     |
|-------------------------------|---------------------------------------------|
| No window                     | "p95 < 400ms" (over what?)                  |
| Average instead of percentile | "avg < 500ms" (hides bimodal distributions) |
| No defense                    | "99.9%" (why this number?)                  |
| Aspirational                  | "99.99%" without multi-region active-active |
| User-mismatched               | "p95 < 5s" for a typing-feedback feature    |

# Activity 1 — SLO Triage

Triads • 35 minutes

8 SLO statements. Identify failures. Rewrite the 5 worst.

- 15 min: triage
- 15 min: rewrite (percentile + window + defense)
- 5 min: identify the most subtle failure



15 + 15 + 5

# Block 2 — Set Your Three SLOs

Latency / Availability / Rate limit



# Error Budget — the Decision Tool

If availability SLO is **99.9%**, error budget is **0.1%** — about **43 minutes** of downtime per 30 days.

- **Inside the budget** → ship features (more risk-tolerant)
- **Near or past the budget** → prioritize reliability (more risk-averse)

**Converts SLO from abstract to load-bearing:** a team that respects its error budget is doing SRE, even if not labeled as such.



# Worked SLOs (FieldPulse reconcile)

```
### SLO – Latency
**Measurement:** Reconcile modal open time, mobile client.
**Target:** p95 ≤ 1000ms, 99% over 30 days.
**Defense:** Dispatchers tap reconcile 30+ times/shift; > 1s
crosses the "I'd rather use paper" threshold (Wk 1
interview Maria, Tier Sheet op-signal #2).
**Verification:** Synthetic transaction in Datadog.

<h3 id="slo--availability">SLO – Availability</h3>
<p><strong>Measurement:</strong> Successful (non-5xx) reconcile submissions.
<strong>Target:</strong> 99.5% over 30 days.
<strong>Error budget:</strong> ~3.6 hrs/month.
<strong>Defense:</strong> Dispatch volume averages 200 reconciles/day; 30-min
outage during peak shift = ~50 affected dispatchers.
<strong>Verification:</strong> Pingdom + 5xx alerting.
<strong>Budget consequence:</strong> If 7-day rolling burns >50% of budget,
we freeze new feature merges and prioritize reliability.</p>
<h3 id="rate-limit-policy">Rate-limit policy</h3>
```

```
<strong>Per-user:</strong> 10 reconcile-submits/sec, burst 30.
<strong>Per-tenant:</strong> 100 reconcile-submits/sec.
<strong>Failure mode:</strong> HTTP 429 with Retry-After header; client
implements exponential backoff with jitter.
<strong>Defense:</strong> Protects against runaway script bug seen in the
2024 dispatcher-tool incident.
<strong>Verification:</strong> Load test before launch; rate-limit cap dashboard.
```

# Activity 2 — Set Your Three SLOs

Triads • 40 minutes

- 15 min: latency SLO (percentile + window + defense)
- 10 min: availability SLO + error-budget translation
- 15 min: rate-limit policy with failure-mode behavior



15 + 10 + 15

# Block 3 — Latency Budget Walk

Does the math fit your Container diagram?



# Per-Hop Latency Reference (rough)

| Hop type                            | Typical range     |
|-------------------------------------|-------------------|
| Network round-trip (LAN → Internet) | 20–80ms           |
| Database read (indexed)             | 5–30ms            |
| Database write (single-row)         | 10–50ms           |
| External API call (sync)            | 50–500ms (varies) |
| Async publish (Kafka/SQS local)     | 1–10ms            |
| JWT validation                      | 1–5ms             |
| TLS handshake (cold)                | 50–200ms          |

**These are starting points.** Use real platform numbers if available.



# The Latency Math

SLO target:  $p95 \leq 1000\text{ms}$

```
<p>Path: Mobile → API GW → Reconcile → Postgres read  
→ Tickets module  
→ Postgres write  
→ Audit publish  
→ response back</p>
```

```
Estimated:  
Mobile-API GW round trip:    80ms  
API GW → Reconcile:         20ms  
Postgres read:  
    30ms  
Tickets module call (sync): 100ms ← dominates  
Postgres write:  
    40ms  
Audit publish (async):       10ms  
Response back:  
    80ms  
-----  
~360ms median  
p95 likely ~700-900ms ✓
```

**If math doesn't fit:** tighter SLO / fewer hops / faster hops / wider percentile.

# Activity 3 — Latency-Budget Walk

Triads • 40 minutes

Annotate your Container diagram with per-hop latency. Sum the path. Decide what gives if it doesn't fit.

- 15 min: annotate per-hop latency
- 10 min: sum the path
- 10 min: if doesn't fit, decide what gives
- 5 min: flag question for engineering



15 + 10 + 10 + 5

# Block 4 — Cross-Review + AI Sanity Check

Three questions; one prompt; honest revision

# The Three Cross-Review Questions

1. **Latency target realistic** given platform precedent?
2. **Availability target realistic** given the team's on-call reality?
3. **Rate-limit policy** protects against a real abuse / accident scenario?

**Author triad listens, captures, decides.** Same Week-3 review discipline.

# AI Sanity Check

```
Role: Senior reliability engineer reviewing a feature's SLO sheet.  
Context: <paste slo="" sheet="" += "" latency="" walk="" rate-limit="" policy="">  
Architecture stance: <paste tcd="" §1="">  
Task: Identify the top 3 reasons the SLOs as written might  
not be sustainable.  
Constraints:  
- Use only the information provided; do not invent platform details  
- Be specific: which SLO, which scenario, which condition  
- Suggest a more realistic alternative for each  
Format: Numbered list – Issue / Scenario / Suggested target.</paste></paste>
```

## Activity 4 — Cross-Review + AI

Triad pairs • 45 minutes • Wrap

- 20 min: cross-review (3 questions)
- 15 min: AI sanity-check + adopt/defer/reject
- 10 min: revise §4 + provenance note



20 + 15 + 10 • 15-min wrap



# Day 4 Wrap

## What we built

- SLI / SLO / SLA / error-budget vocabulary
- Three SLOs (latency / availability / rate)
- Latency-budget walk through Container diagram
- Error-budget consequence stated
- TCD §4 drafted

## What opens tomorrow

- **Technical trade-offs & constraints**
- Friday: assemble TCD §§5–6
- Stakeholder + sign-off matrix
- Cross-triad review of the full TCD

**Bring to Day 5:** Your TCD §§1–4. Tomorrow integrates everything into the final TCD.

# Questions & Reflection

Which of your SLOs would you fold under pressure from a stakeholder asking for "just a bit more"?